

KAI YUE

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EDUCATION

Ph.D. in Electrical Engineering (GPA: 3.97/4.0)

Aug. 2021 – May 2025 (expected)

North Carolina State University (NC State)

- Advisors: Dr. Chau-Wai Wong and Dr. Huaiyu Dai

B.E. in Electronic Information Engineering (GPA: 3.92/4.0)

Sept. 2016 – Jul. 2020

University of Science and Technology of China (USTC)

- Advisors: Dr. Dong Liu and Dr. Feng Wu

PUBLICATIONS

[C1] **Kai Yue**, Richeng Jin, Ryan Pilgrim, Chau-Wai Wong, Dror Baron, and Huaiyu Dai. Neural tangent kernel empowered federated learning. *International Conference on Machine Learning (ICML'22)*.

[C2] **Kai Yue**, Richeng Jin, Chau-Wai Wong, Dror Baron, and Huaiyu Dai. Gradient obfuscation gives a false sense of security in federated learning. *Accepted by USENIX Security Symposium, 2023*.

[J 1] **Kai Yue**, Richeng Jin, Chau-Wai Wong, and Huaiyu Dai. Communication-efficient federated learning via predictive coding. *IEEE Journal of Selected Topics in Signal Processing, 2022*.

[J 2] **Kai Yue**, Richeng Jin, Chau-Wai Wong, and Huaiyu Dai. Federated learning via plurality vote. *IEEE Transactions on Neural Networks and Learning Systems, 2022*.

RESEARCH EXPERIENCES

NC State University (Research Assistant)

Aug. 2021 – Present

Advisors: Dr. Chau-Wai Wong & Dr. Huaiyu Dai

- Handled statistical heterogeneity by designing neural tangent kernel empowered federated learning [C1]
- Studied a new attack framework and the privacy leakage problem in federated learning [C2]
- Reduced communication overhead by 99% using predictive coding for federated learning [J 1]
- Improved model efficiency and robustness by designing federated learning via plurality vote [J 2]

University of California, Los Angeles (Summer Intern)

Jul. 2019 – Sept. 2019

Advisors: Dr. Yixin Zhu & Dr. Song-Chun Zhu

- Improved a robot operating system (ROS) based glove for real-time hand pose tracking in the MuJoCo physics engine, allowing users to interact with the engine more naturally and smoothly
- Designed latent representation based hand pose reconstruction algorithms using autoencoders

USTC (Undergraduate Researcher)

Aug. 2018 – Jan. 2020

Advisors: Dr. Dong Liu & Dr. Feng Wu

- Worked in a team of four and took the lead to design reference frame generation algorithms
- Saved 28% of Bjøntegaard Delta bit rate (BD-Rate) for underwater video coding
- Won the second place in Water Bottom Video Challenge (WBVC) 2018

PROJECTS

Online Intelligent Tutoring Platform Design

Nov. 2018 – Feb. 2019

- Worked in a team of five and incorporated neural network tools into an online tutoring platform
- Won the third place in International Youth Leadership Finance Summit (IYLFS) 2019 competition

Failure Prediction on Wind Turbine Systems

Jun. 2018 – Jul. 2018

- Worked in a team of four under supervision of Dr. Hongzhi Wang
- Reduced the running time by more than 50% by improving the efficiency of K-means algorithm

AWARDS AND SCHOLARSHIPS

USENIX Security 2023 Student Grant	2023
Graduate Award, NC State University	2023
ICML 2022 Participation Grant	2022
Summer Graduate Fellowship, NC State University	2022
Graduate Merit Award, NC State University	2021
Honorable Title of Excellent Undergraduate Students, USTC	2020
UCLA–CSST Scholarship, UCLA	2019
Third Place for online intelligent tutoring platform design, IYLFS 2019 competition	2019
Second Place for bit rate savings in underwater video coding, WBVC 2018 competition	2018
Mr. Tang-Lixin’s Scholarship (known as the “Academy Award of scholarships” in China)	2018
National Scholarship, the Ministry of Education of China (Top 1%)	2018

PROFESSIONAL SERVICE

Conference Reviewer

- CVPR 2023, ICCV 2023, CVPR 2022, ECCV 2022, ICASSP 2022

Representative of Talent Program in Artificial Intelligence, USTC

- Organized off-campus projects at iFlytek Co., Ltd for students in the program 2019

TEACHING EXPERIENCE

Teaching Assistant, NC State

- ECE 411 Introduction to Machine Learning Fall 2022
- ECE 209 Computer Systems Programming Fall 2021, Spring 2022

GEARS Research Program Mentor, NC State

- Guided undergraduate students to conduct federated learning research Spring 2022, Summer 2022

TECHNICAL SKILLS

Programming	Python, R, C, C++, CUDA, Java, $\text{\LaTeX}$
Research	Federated Learning, Machine Learning, Statistical Learning, Image Processing